



KNOX GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT

2002
YEARLY EXAMINATION

Mathematics Extension 1

Year 11

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- Board-approved calculators may be used
- All necessary working should be shown in every question

Total marks (84)

- Attempt Questions 1–7
- All questions are of equal value
- Use a **SEPARATE** Writing Booklet for each question

If no attempt is made on a question, write the Question Number and your Student Number in the appropriate place on the cover, and write NIL ATTEMPT in the Mark Column.

HAND IN THIS BOOKLET.

NAME: _____

TEACHER: _____

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Total marks (84)

Attempt questions 1 – 7

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks) Use a SEPARATE writing booklet.		Marks
(a)	Solve $\frac{3}{x-2} > 4$.	3
(b)	Factorise $a^2 - 4b^2 + 2a + 4b$.	2
(c)	Solve $\sin 2\theta = \frac{\sqrt{3}}{2}$ for $0^\circ \leq \theta \leq 360^\circ$.	2
(d)	If $x = 4 + 2\sqrt{3}$, simplify $x - \frac{1}{x}$.	3
(e)	Using the fact that $(\sqrt{x} + \sqrt{y})^2 = x + y + 2\sqrt{xy}$, express $11 + 4\sqrt{6}$ in the form $(\sqrt{x} + \sqrt{y})^2$.	2

Question 2 (12 marks) Use a SEPARATE writing booklet.

(a)	$f(x) = x^2 + 3x - 2$. Simplify $\frac{f(h) - f(-h)}{2h}$.	3
(b)	Find the inverse function of the function $y = \frac{2x-3}{x-1}$ in the form $y = f(x)$.	3
(c)	Solve $x^2 > 9x$.	3
(d)	Solve $ x-2 = 2x+4$.	3

Question 3 (12 marks) Use a SEPARATE writing booklet.

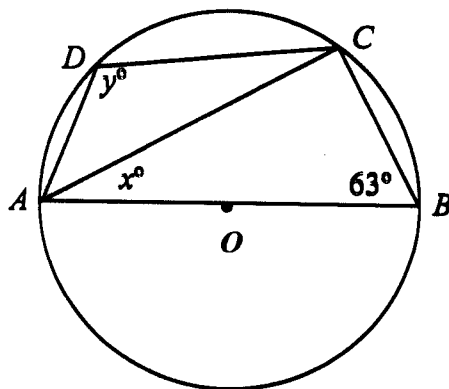
Marks

- (a) By drawing appropriate straight lines on the same axes as the graph of $y = 2^x$, determine the number of solutions for each of the following equations: 3
- (i) $2^x = x + 2$
- (ii) $2^x = x - 2$
- (iii) $2^x = 2 - x$
- (b) Find $\lim_{x \rightarrow 2} \frac{x^2 - 4}{2x - 4}$ 2
- (c) If $\sin \theta = p$, where θ is acute, find expressions for $\cos \theta$ and $\tan \theta$ in terms of p . 2
- (d) Solve for $0^\circ \leq \theta \leq 360^\circ$, to the nearest degree: $4 \sec^2 \theta - \tan \theta = 7$. 5

Question 4 (12 marks) Use a SEPARATE writing booklet.

- (a) Find the coordinates of the point that divides the join of $(-3, 2)$ and $(5, 7)$ **externally** in the ratio $1 : 3$. 3
- (b) Find the perpendicular distance between the pair of parallel lines $3x - 4y + 12 = 0$ and $3x - 4y - 6 = 0$. 3
- [Hint: Find the coordinates of a point on one of the lines.]
- (c) If $f(x) = \frac{x-1}{\sqrt{2x+1}}$, show that $f'(x) = \frac{x+2}{\sqrt{(2x+1)^3}}$. 4

- (d) 2



$ABCD$ is a cyclic quadrilateral, and AB is the diameter of the circle, with centre O .
 $\angle ABC = 63^\circ$, $\angle CAB = x^\circ$, and $\angle ADC = y^\circ$.
 Find the values of x and y , giving reasons.

Question 5 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Find the values of a, b, c given that $x^2 + x + 1 \equiv a(x-1)^2 + b(x-1) + c$.

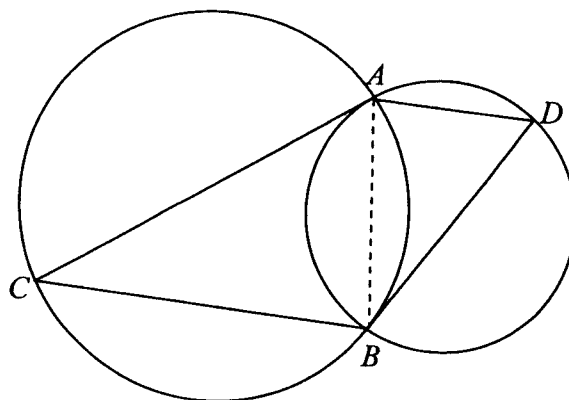
3

(b) Find the maximum value of the function $f(x) = 8 + 4x - x^2$.

2

(c)

3



Two circles intersect at A and B . AC is a tangent to one circle at A and BD is a tangent to the other circle at B .

Copy or trace the diagram into your writing booklet.

Prove that AD is parallel to CB .

(d) (i) Sketch the graphs of $y = |x^2 - 2|$ and $y = 3$ on the same number plane.

2

(ii) Write down the inequalities that simultaneously represent the region bounded by the two graphs.

2

Question 6 (12 marks) Use a SEPARATE writing booklet.

(a) Find the value of m if the quadratic equation $x^2 + mx + (m + 2) = 0$:

(i) has one root equal to 3.

2

(ii) has its roots reciprocals of each other.

2

(b) Prove that $x^2 + (k - 3)x - k = 0$ has real roots for all real values of k .

3

(c) (i) Find the equation of the tangent to the curve $y = x^2$ at the point (a, a^2) on the curve.

2

(ii) Find the values of a if this tangent passes through the point $(2, -5)$.

1

(iii) Hence find the equations of the two tangents to the curve $y = x^2$ that pass through the point $(2, -5)$.

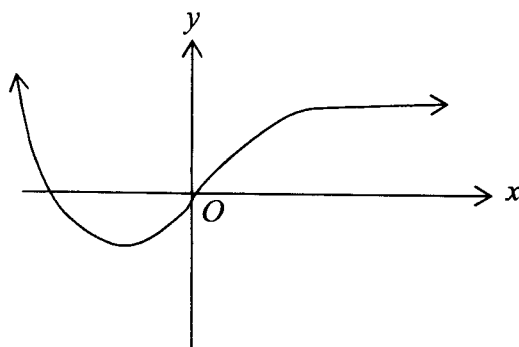
2

Question 7 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Find the coordinates of the points on the curve $y = x(x - 2)^3$ where the tangent to the curve is horizontal (i.e. has a gradient of zero). 4

- (b) 2

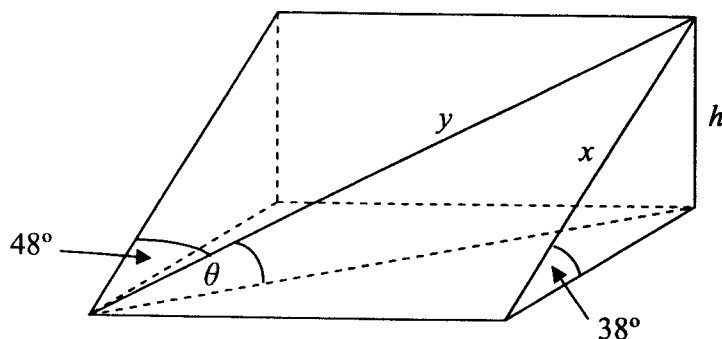


The diagram shows the graph of $y = f(x)$.

Copy or trace this graph into your Writing Booklet.

On the same set of axes draw the graph of $y = f'(x)$.

- (c)



A hill is inclined at an angle of 38° to the horizontal. A road on the hill makes an angle of 48° with a line of greatest slope.

- (i) Show that the angle of inclination of the road to the horizontal (θ) is given by: 2
 $\sin \theta = \sin 38^\circ \cos 48^\circ$.
- (ii) Hence find the value of θ . 1

- (d) Sketch the graph of $|xy| \geq 4$ on a number plane. 3

End of Examination

1(a) $\frac{x-3}{x-2} > 4$

$3(x-2) > 4(x-2)^2$

$3x-6 > 4x^2-16x+16$

$4x^2-19x+22 < 0$

$(4x-11)(x-2) < 0$

$2 < x < 2\frac{1}{4}$

③

(b) $a^2-4b^2+2a+4b$

$= (a-2b)(a+2b) + 2(a+2b)$

$= (a+2b)(a-2b+2)$

②

(c) $\sin 2\theta = \frac{\sqrt{3}}{2}$

$2\theta = 60^\circ, 120^\circ, 420^\circ, 480^\circ$

$\theta = 30^\circ, 60^\circ, 210^\circ, 240^\circ$

②

(d) $x = 4 + 2\sqrt{3}$

$\frac{1}{x} = \frac{1}{4+2\sqrt{3}} \times \frac{4-2\sqrt{3}}{4-2\sqrt{3}}$

$= \frac{4-2\sqrt{3}}{4}$

$x - \frac{1}{x} = 4 + 2\sqrt{3} - \frac{4-2\sqrt{3}}{4}$

$= \frac{16+8\sqrt{3}-(4-2\sqrt{3})}{4}$

$= \frac{12+10\sqrt{3}}{4}$

OR $\frac{6+5\sqrt{3}}{2}$

③

(i) $11+4\sqrt{6} = 11+2\sqrt{24}$

$x+y = 11, xy = 24$

$x = 8, y = 3$

$\therefore 11+4\sqrt{6} = (\sqrt{8}+\sqrt{3})^2$

②

2(a) $f(x) = x^2+3x-2$

$\frac{f(4)-f(-4)}{2h} = \frac{(4^2+3\cdot 4-2)-((-4)^2-3\cdot 4-2)}{2h}$

$= \frac{64}{2h}$

$= 3$

③

(b) $y = \frac{2x-3}{x-1}$

Inverse is $x = \frac{2y-3}{y-1}$

$xy-x = 2y-3$

$xy-2y = x-3$

$y(x-2) = x-3$

$y = \frac{x-3}{x-2}$

③

(c)

$x^2 > 9x$

$x^2-9x > 0$

$x(x-9) > 0$

$x < 0, x > 9$

③

(d)

$|x-2| = 2x+4$

Note: $2x+4 \geq 0, \therefore x \geq -2$.

$x-2 = 2x+4$ or $x-2 = -(2x+4)$

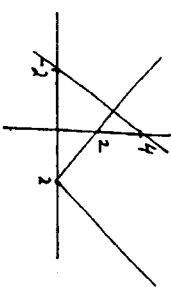
$-6 = x$ or $3x = -2$

$x = -6$ or $x = -\frac{2}{3}$

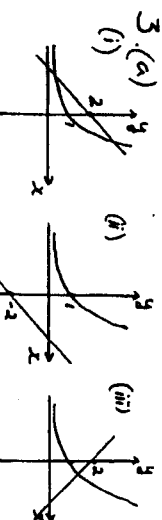
But $x \geq -2 \therefore x = -\frac{2}{3}$ is only solution

OR Test $x = -6, x = -\frac{2}{3}$ into eqn

OR Draw graphs and solve:



③



2 solutions 0 solutions 1 solution

③

(b) $\lim_{x \rightarrow 2} \frac{x^2-4}{2x-4} = \lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{2(x-2)}$

$= \lim_{x \rightarrow 2} \frac{x+2}{2}$

$= \frac{4}{2}$

$= 2$

②

(c)

$\sin \theta = p$

$\cos^2 \theta = 1 - \sin^2 \theta$

$\cos \theta = \sqrt{1-p^2}$

$\tan \theta = \frac{\sin \theta}{\cos \theta}$

$= \frac{p}{\sqrt{1-p^2}}$

②

(d)

$4 \sec^2 \theta - \tan \theta = 7$

$4(\tan^2 \theta + 1) - \tan \theta - 7 = 0$

$4 \tan^2 \theta - \tan \theta - 3 = 0$

$(4 \tan \theta + 3)(\tan \theta - 1) = 0$

$\tan \theta = -\frac{3}{4}$ or $\tan \theta = 1$

$\theta = 180^\circ - 37^\circ, 360^\circ - 37^\circ, 45^\circ, 180^\circ + 45^\circ$

$= 143^\circ, 323^\circ, 45^\circ, 225^\circ$

⑤

4(a) $\begin{pmatrix} x_1 & y_1 & x_2 & y_2 \\ (-3, 2) & (5, 7) & (-1, 3) \end{pmatrix}$

$\begin{pmatrix} (-1) \times 5 + 3 \times (-3) & (-1) \times 7 + 3 \times 2 \\ -1+3 & -1+3 \end{pmatrix}$

$\begin{pmatrix} -7 & -2 \end{pmatrix}$

③

(b) $3x-4y+12=0, 3x-4y-6=0$

(3,0) lies on $3x-4y-6=0$

Perp. dist. = $\frac{|3 \times 2 - 4 \times 0 + 12|}{\sqrt{3^2+4^2}}$

$= \frac{18}{5}$ units

③

(c) $f(x) = \frac{x-1}{\sqrt{2x+1}}$

$f'(x) = \frac{(\sqrt{2x+1} \times 1 - (x-1) \times \frac{1}{2} \sqrt{2x+1})}{2x+1}$

$= \frac{\sqrt{2x+1} - \frac{x-1}{2} \sqrt{2x+1}}{2x+1} \times \frac{\sqrt{2x+1}}{\sqrt{2x+1}}$

$= \frac{2x+1 - (x-1)}{2x+1}$

$= \frac{x+2}{2x+1}$

$= \frac{x+2}{\sqrt{(2x+1)^3}}$

④

(d) $\angle ACB = 90^\circ$ (angle in semi-circle)

$\therefore x = 180 - (90 + 63)$ (angle sum of triangle)

$= 27$

$y = 180 - 63$

(opposite angles of cyclic quad.)

②

5(a) $x^2 + x + 1 = a(x-1)^2 + b(x-1) + c$
 $= ax^2 + (-2a+b)x + a-b+c$

$\therefore a = 1$
 $-2a + b = 1 \therefore b = 3$
 $a - b + c = 1 \therefore c = 3$

(3)

(b) $f(x) = 8 + 4x - x^2$

EITHER: Axis of symmetry: $x = \frac{-b}{2a} = 2$

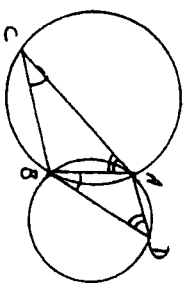
$f(2) = 8 + 8 - 4 = 12$

OR: $f(x) = -(x^2 - 4x + 4) + 8 + 4$
 $= 12 - (x-2)^2$

Maximum value is 12

(2)

(c)



$\angle ADB = \angle ACB$ (alt. seg. thm)

$\angle CAB = \angle CDB$ (alt. seg. thm)

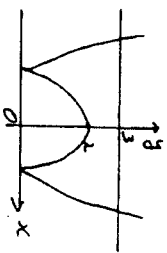
Since two pairs of angles of a triangle are equal, the third pair are equal

$\therefore \angle ABC = \angle BAD$

$\therefore AD \parallel CB$ (corresponding angles are equal.)

(3)

(i)



(2)

(ii)

$y \leq 3, y \geq |x^2 - 2|$

(2)

6(a) $x^2 + mx + (m+2) = 0$

(i) one root 3: $9 + 3m + (m+2) = 0$

$4m = -11$

$m = -2\frac{1}{4}$

(2)

(ii) reciprocals: Product = 1

$m + 2 = 1$

$m = -1$

(2)

(b) $x^2 + (k-3)x - k = 0$

$\Delta = (k-3)^2 - 4 \times 1 \times (-k)$

$= k^2 - 6k + 9 + 4k$

$= k^2 - 2k + 9$

$= (k-1)^2 + 8$

$\Delta > 0$ for all k

$\therefore$ roots are real for all real k .

(3)

(c) (i) $y = x^2$

$\frac{dy}{dx} = 2x$

At (a, a^2) $\frac{dy}{dx} = 2a$

Tangent: $y - a^2 = 2a(x - a)$

$y - a^2 = 2ax - 2a^2$

$y = 2ax - a^2$

(3)

(ii) Subst. $x = 2, y = -5$:

$-5 = 2 \times a \times 2 - a^2$

$a^2 - 4a - 5 = 0$

$(a-5)(a+1) = 0$

$a = 5$ or $a = -1$

(1)

(iii) $a = 5$: $y = 10x - 25$

$a = -1$: $y = -2x - 1$

(2)

7(a) $y = x(x-2)^3$

$\frac{dy}{dx} = (x-2)^3 \times 1 + x \times 3(x-2)^2$

$= (x-2)^3 + 3x(x-2)^2$

$= (x-2)^2 [(x-2) + 3x]$

$= (x-2)^2 (4x-2)$

Curve is horizontal when $\frac{dy}{dx} = 0$,

i.e. $x = 2, x = \frac{1}{2}$.

When $x = 2, y = 0$.

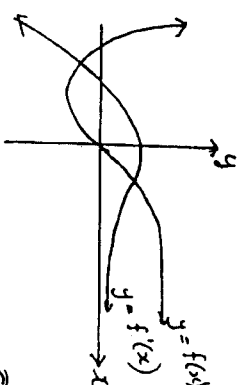
When $x = \frac{1}{2}, y = \frac{1}{2} \times \left(-\frac{5}{2}\right)^3$

$= -\frac{25}{8}$

Points are $(2, 0), \left(\frac{1}{2}, -\frac{25}{8}\right)$

(4)

(b).



(2)

(c) (i) $\sin 38^\circ \cos 48^\circ = \frac{1}{2} \times \frac{1}{2}$

$= \frac{1}{4}$

$= \sin \theta$

(2)

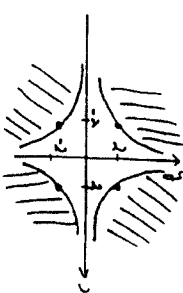
(ii) $\sin \theta = \sin 38^\circ \cos 48^\circ$

$\therefore 0.4119 \dots$

$\theta = 24^\circ 20'$

(1)

(d) $|xy| \geq 4$ i.e. $xy \geq 4$ or $xy \leq -4$



(3)